Vera Molnar's EXPLORATION OF THE UNIMAGINABLE

Rachel R. Roe-Dale

The once hidden stories of female mathematicians such as Katherine Johnson, Dorothy Vaughan, and Mary Jackson, who worked as computers during the 1960s race to space, have recently been brought to light through Margot Lee Shetterly's novel and the subsequent film *Hidden Figures* (see *Math Horizons*, February 2017). Vera Molnar is another female pioneer in the computing era who deserves more exposure.

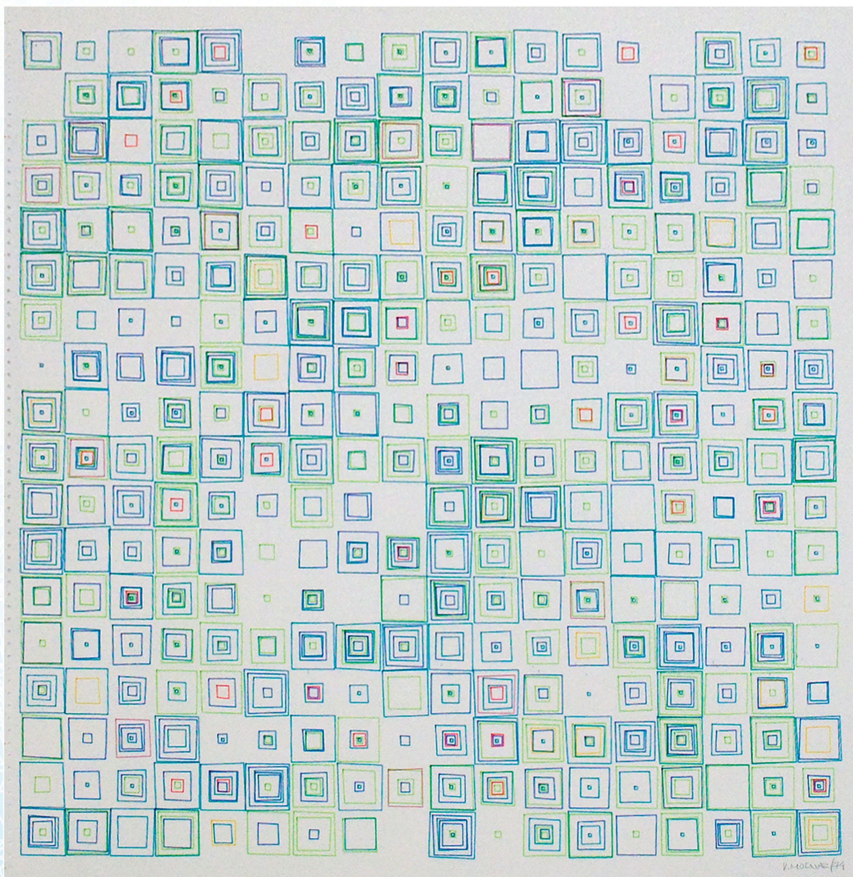
Molnar is a Hungarian artist whose work is rising from obscurity in the United States partially as a result of recent exhibitions such as *Thinking Machines: Art and Design in the Computer Age, 1959–1989* at the Museum of Modern Art in New York City and *Vera Molnar: Drawings 1949–1986* and *Regarding the Infinite* at Senior & Shopmaker's gallery in New York City. Like her computing contemporaries, Molnar was a pioneer in the field she helped shape and develop—computer-aided art and design.

I first became interested in Molnar and her story when we showed two examples of her series *(Des) Ordres* (see figure 1) at the Frances Young Tang Teaching Museum and Art Gallery in the exhibition, *Sixfold Symmetry: Pattern in Art and Science* (*http://bit.ly/VeraMolnar*), which I curated with my colleague Rachel Seligman, Assistant Director for Curatorial Affairs and Malloy Curator at the Tang Museum.

Vera Molnar's *Machine Imaginaire*

Vera Molnar was born in Budapest in 1924, and at 95, Molnar is still producing work. She lives in Paris, where she spent most of her professional

Figure 1. Vera Molnar. Detail of *(Des) Ordres*, 1974. Computer graphic on Benson plotter paper, 40 1/4 × 30 3/8 inches.



Collection of the artist; courtesy Senior & Shopmaker Gallery, New York.

career. Trained at the Budapest College of Fine Arts, Molnar moved away from traditional art, and her earliest works, typically oil paintings, were abstract paintings that often featured geometric shapes characteristic to those in the concrete art movement. In 1960, Molnar was a founding member of the Groupe de Recherche d'Art Visuel (Visual Arts Research Group).

Around this time—in the early days of the computer's development—machine access was restricted to scientific research and military purposes. Anticipating the computer's use in producing and developing her work, Molnar relied on what she referred to as her "*machine imaginaire* (imaginary machine)." Using her physical skills, she executed her algorithms for production by deliberate, manual iteration of specific algorithmic processes, a process akin to that used by her contemporary human computers—Johnson, Vaughan, and Jackson—as they manually performed complex mathematical calculations. For Molnar, that meant dividing her working surface (usually paper) into a grid and following her own self-imposed guidelines and distributions of parameters such as color, shape, and slope to fill the grid.

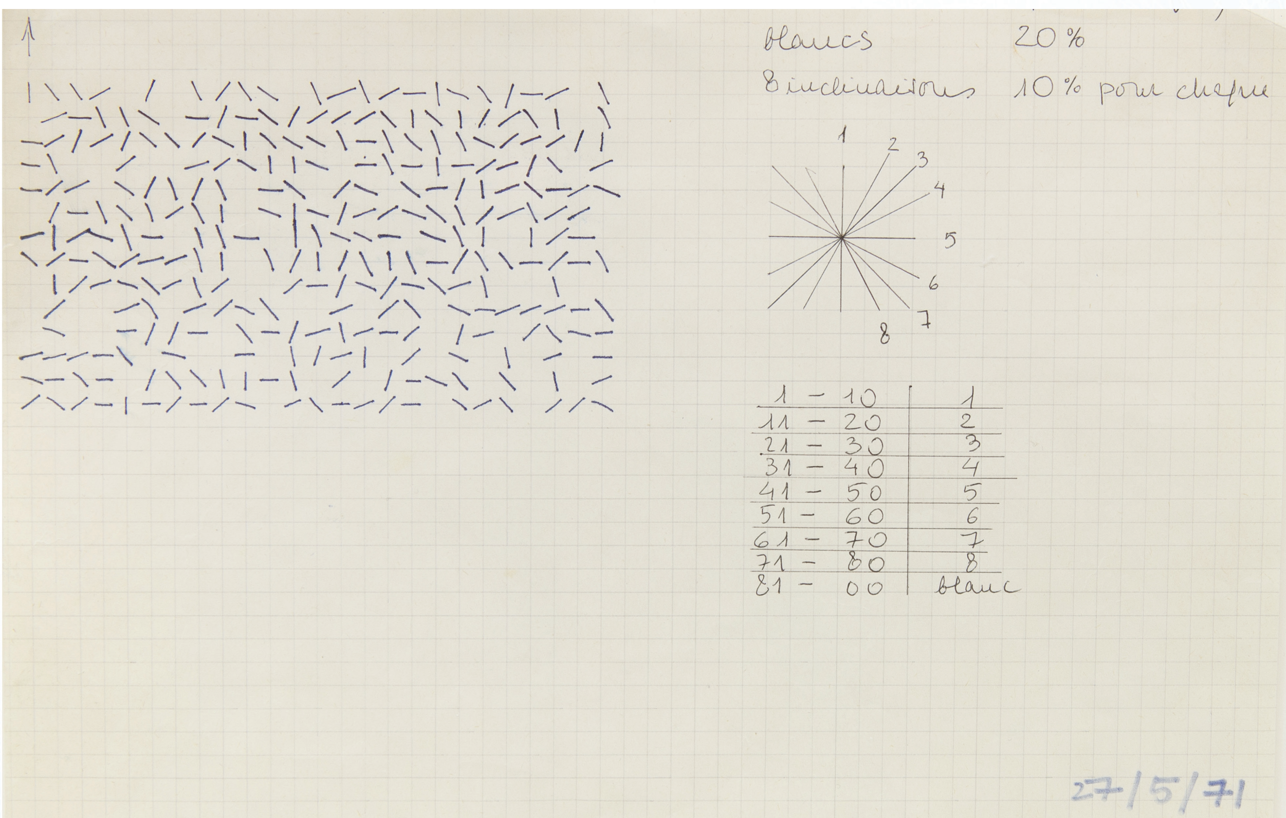
In *Distribution Aleatoire de 4 Elements (pour prog. ordinateur)*, shown in figure 2, Molnar fills the grid with line segments of differing slopes. Her notes describing the distribution of slopes are included to the right of the piece. Mathematicians will likely identify with this work, perhaps thinking about the function that generated this field-like plot.

Computer-Aided Art and Design

Following experimentation with her *machine imaginaire*, Molnar obtained access to a Bull Series 300 computer and a Benson plotter at Bull Information Systems Research Center in Paris in 1968. She taught herself to program in FORTRAN and BASIC, and she used the computer to further her combinatorial exploration by varying parameters such as shape, scale, line thickness, and color.

Automating the design process with a computer allowed Molnar to create work more quickly and increase throughput. More importantly, the computer-aided approach expanded Molnar's

Figure 2. Vera Molnar. *Distribution Aleatoire de 4 Elements (pour progr. ordinateur)*, 1971. Ink and graphite on paper, 7 1/2 x 10 1/2 inches.



Collection of the artist; courtesy Senior & Shopmaker Gallery, New York.

aesthetic investigation by allowing her to create more parameter combinations than she could have done previously due to the physical limitations of production by hand.

Molnar described her experience with the computer in a statement included in a recent exhibition at the Digital Art Museum Gallery in Berlin. She explained,

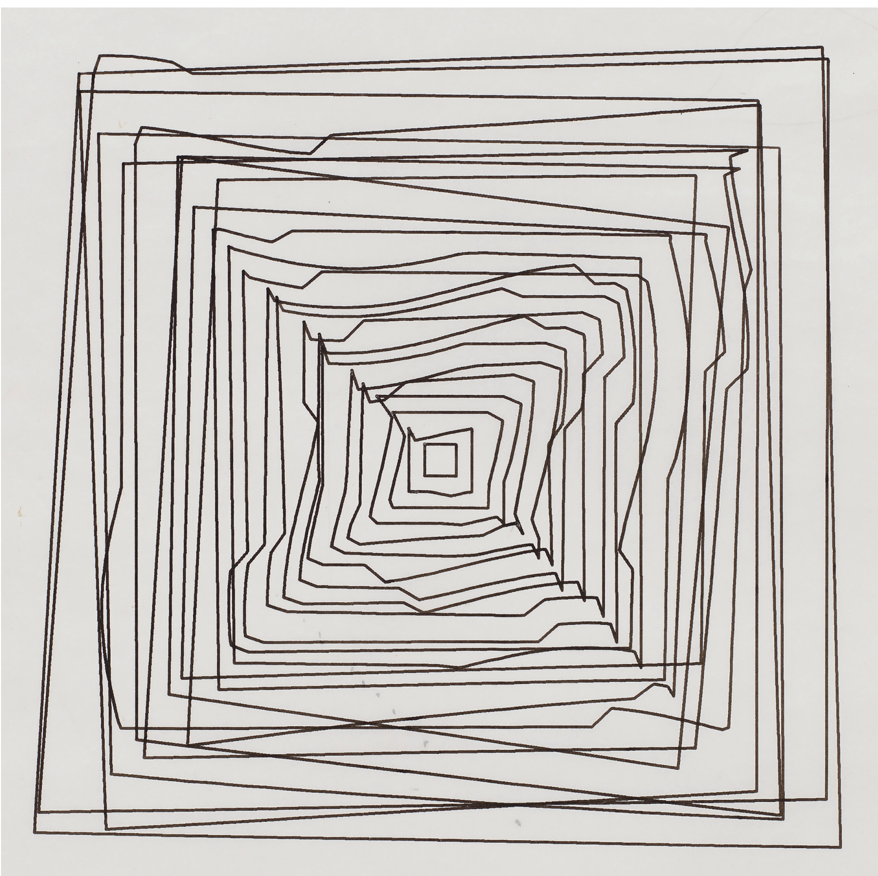
> I use a computer to combine the forms because I hope that the assistance of this tool will permit me to go beyond the bounds of learning, cultural heritage, environment; in short: of the social thing, which we must consider to be our second nature. Because of its huge capacity for combination, the computer permits systematic investigation of the field of possibilities in the visual world, helping the painter to clear his brain of mental/cultural 'ready-mades' and in enabling him to produce combinations of forms never seen before, either in nature, or in museums, to create unimaginable images.

Using the computer gave Molnar the ability to explore the unimaginable.

After using the Bull, Molnar moved on to an IBM 2250 at the Université de Paris I, Sorbonne in Orsay, which she used from 1969 to 1976. She used a Tektronix machine at the Centre George Pompidou from 1977 to 1979 before moving to a personal computer in 1980.

When she produced her designs with algorithms coded on punch cards, Molnar often had to wait several days for her code to be executed and the plotters to produce her coded images. Later advances in technology allowed her to review her art on a screen mid-algorithm as the work was evolving. Visualizing specific combinations of parameters on the screen allowed her to capture in situ the interactions of the parameters. She dubbed this method of production the "conversational method." Only with the assistance of the computer was this technique possible.

Figure 3. Vera Molnar. Detail of *Hypertransformation* of 20 *concentric squares*, 1974. Computer drawing on Benson plotter paper, 20 1/4 × 14 1/8 inches.

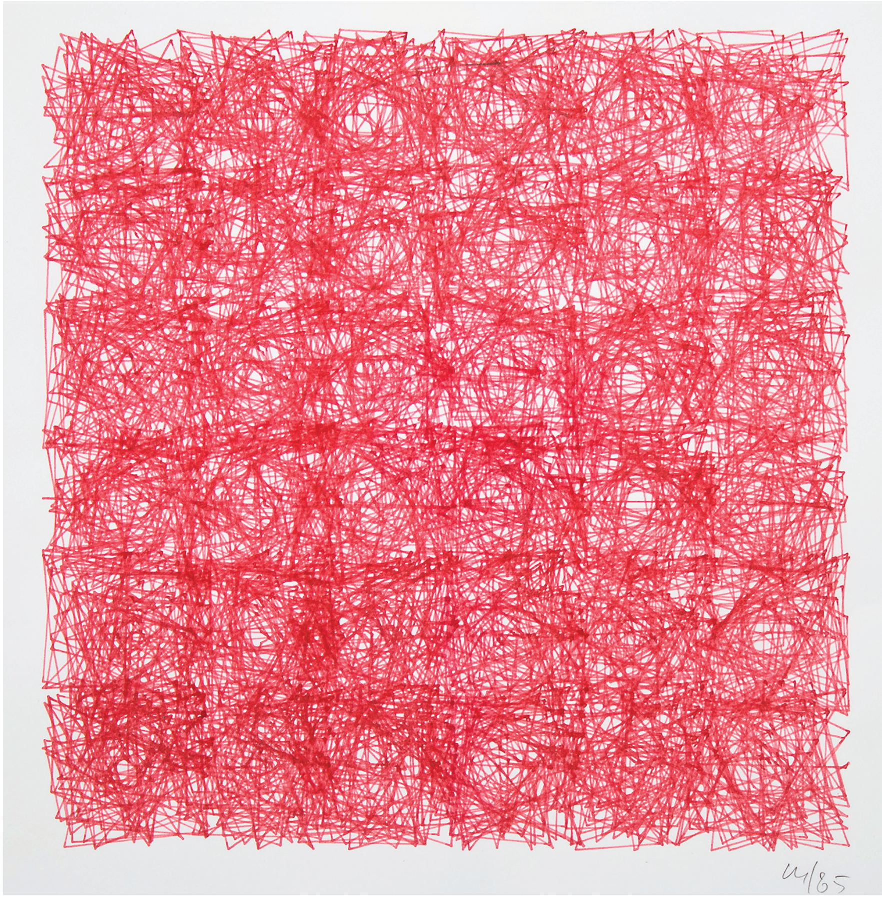


Collection of the artist; courtesy Senior & Shopmaker Gallery, New York.

In one series, Molnar started with 20 concentric squares. She described the evolution of this work in a 1975 article "Toward Aesthetic Guidelines for Paintings with the Aid of a Computer" for *Leonardo*. Starting with an initial idea (what we might refer to as a seed), Molnar followed a "series of small, probing steps" in which she altered specific parameters in a systematic way. She paused after each alteration to compare successive images, taking note of whether the trend was pleasing. For this series, Molnar experimented with replacing the edges of the squares with curved segments from parabolas, hyperbolas, and sine curves. Relying on her intuition to realize beauty, Molnar ultimately settled on replacing some of the edges with sine curve segments to produce *Hypertransformation of 20 Concentric Squares*, shown in figure 3.

After determining an optimal combination and arrangement of parameters that realized her original idea, Molnar often experimented with disorder, introducing small perturbations in the design parameters. At this point of synthesis, Molnar described these anomalies as adding "one percent disorder." While not mathematically quantifiable, these perturbations represented Molnar's investigations into randomness, a process mathematicians might model with a stochastic system. *(Des) Ordres* (figure 1) is an example of work generated in this manner. Another more recent work, *Structure de quadrilatères (square structures*), shown in figure 4, also explores these themes.

Figure 4. Vera Molnar *Structure de quadrilatères (square structures)*, 1985. Computer graphic on Benson Calcomp paper roll 11 5/8 x 11 5/8 inches.



Private Collection, London; Courtesy of Senior & Shopmaker Gallery, New York. © Vera Molnar.

Molnar was one of the pioneers of computer-generated art and one of the first artists to translate algorithmic thinking into concrete, artistic output. Her work illustrates mathematical and algorithmic concepts and provides a chronology of the evolution of computer-aided art and access to computers.

Using a computer in the design process was revolutionary. As with many revolutionaries, it has taken time for Molnar to receive the credit she deserves. Recent exhibitions, interest, and advances in digital media have helped uncover work that was once hidden, and we hope that this trend of exploring the role of mathematics, algorithms, and computer science in artistic development and production will continue.

The artwork in this article has been cropped to show the artistic details. The reader can see the uncropped artwork art by visiting *maa.org/mathhorizons/supplemental.htm*.

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10.1080/10724117.2019.1568081

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1	8	6	4	7	3	9	5	2
7	2	9	1	6	5	3	8	4
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Solution to Square or Prime Differencedoku (see page 2).